

Exercise: 9.3

Solve the following Questions
Q No: 01

If $u = x - y^2$, $x = 2r - 3s + 4$, $y = -r + 8s - 5$.

and find $\frac{\partial u}{\partial r}$ and $\frac{\partial u}{\partial s}$

Sol:

Main formula

$$\therefore \frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial r} + \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial r} \dots$$

$u = x - y^2$, $x = 2r - 3s + 4$, $y = -r + 8s - 5$
formula

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial r}$$

$$= \frac{\partial}{\partial x} (x - y^2) \cdot \frac{\partial}{\partial r} (2r - 3s + 4) + \frac{\partial}{\partial y} (x - y^2) \cdot \frac{\partial}{\partial r} (-r + 8s - 5)$$

$$= 1 \cdot 2 + (-2y) \cdot (-1)$$

$$\frac{\partial u}{\partial r} = 2 + 2y$$

$$\frac{\partial u}{\partial s} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial s}$$

$$= \frac{\partial}{\partial x} (x - y^2) \cdot \frac{\partial}{\partial s} (x - 2r - 3s + 4) + \frac{\partial}{\partial y} (x - y^2) \cdot \frac{\partial}{\partial s} (-r + 8s - 5)$$

$$= 1 \cdot (-3) + (-2y) \cdot (8)$$

$$\frac{\partial u}{\partial x} = -3 - 16y$$

1) $z = \frac{\cos y}{x}$, $x = u^2 - v$, $y = e^v$

find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$

Sol:

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$= \frac{\partial}{\partial x} \left(\frac{\cos y}{x} \right) \cdot \frac{\partial}{\partial u} (u^2 - v) + \frac{\partial}{\partial y} \left(\frac{\cos y}{x} \right) \cdot \frac{\partial}{\partial u} (e^v)$$

$$\cos y \cdot \frac{\partial}{\partial x} \left(\frac{1}{x} \right) = \frac{-\cos y}{x^2} \cdot 2u + \left(-\frac{1}{x} \sin y \right) \cdot 0$$

$$\frac{-\cos y}{x^2} = \frac{-2u \cdot \cos y}{x^2}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$$

$$= \frac{\partial}{\partial x} \left(\frac{\cos y}{x} \right) \cdot \frac{\partial}{\partial v} (u^2 - v) + \frac{\partial}{\partial y} \left(\frac{\cos y}{x} \right) \cdot \frac{\partial}{\partial v} (e^v)$$

$$= \frac{-\cos y}{x^2} \cdot (-1) + \frac{-\sin y}{x} \cdot e^v$$

$$e^v = y \text{ (given)}$$

$$= \frac{\cos y}{x} - \frac{e^v \cdot \sin y}{x}$$

$$\frac{\cos y}{x} - \frac{y \cdot \sin y}{x}$$

$$= \frac{\cos y - y \cdot \sin y}{x}$$

$$\sin xy - e^{xy} - x^2y = 0$$

$$\therefore \frac{dy}{dx} = -\frac{f_x}{f_y}$$

Sol:

$$f(x, y) = \sin xy - e^{xy} - x^2y \quad \text{--- (1)}$$

partially diff (1) w.r.t "x"

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} (\sin xy - e^{xy} - x^2y)$$

$$= \cos xy \cdot (y) - e^{xy} (y) - 2xy$$

$$f_x = y \cos xy - ye^{xy} - 2xy$$

partially diff (1) w.r.t "y"

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (\sin xy - e^{xy} - x^2y)$$

$$= \cos xy (x) - e^{xy} (x) - x^2$$

$$f_y = x \cos xy - xe^{xy} - x^2$$

$$\therefore \frac{dy}{dx} = -\frac{f_x}{f_y}$$

$$\frac{dy}{dx} = - \left(\frac{y \cos xy - ye^{xy} - 2xy}{x \cos xy - xe^{xy} - x^2} \right)$$

$$\frac{dy}{dx} = \frac{y \cos xy - ye^{xy} - 2xy}{x(-\cos xy + e^{xy} - x)}$$

$$(4) \quad 3(x^2 + y^2)^2 = 25(x^2 - y^2)$$

Sol:

$$b(x, y) = 3(x^2 + y^2)^2 - 25(x^2 - y^2) \quad \text{--- (1)}$$

partially diff w.r.t "x"

$$\frac{\partial b}{\partial x} = \frac{\partial}{\partial x} (3(x^2 + y^2)^2 - 25(x^2 - y^2))$$

$$= 3 \cdot [2(x^2 + y^2)^{2-1} (2x)] - 25(2x)$$

$$b_x = 12x(x^2 + y^2) - 50x$$

partially diff w.r.t "y"

$$\frac{\partial b}{\partial y} = \frac{\partial}{\partial y} (3(x^2 + y^2)^2 - 25(x^2 - y^2))$$

$$= 3(2(x^2 + y^2)(2y)) - 25(-2y)$$

$$b_y = 12y(x^2 + y^2) + 50y$$

$$\therefore \frac{dy}{dx} = \frac{-b_x}{b_y}$$

$$\frac{-b_x}{b_y} = \frac{-[12x(x^2 + y^2) - 50x]}{12y(x^2 + y^2) + 50y}$$

$$= \frac{-12x(x^2 + y^2) + 50x}{12y(x^2 + y^2) + 50y}$$

$$= \frac{2(25x - 6x(x^2 + y^2))}{2(6y(x^2 + y^2) + 25y)}$$

$$\frac{dy}{dx} = \frac{25x - 6x(x^2 + y^2)}{6y(x^2 + y^2) + 25y}$$

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$$b(x, y) = x^y - y^x = 0$$

Sol:

$$b(x, y) = x^y - y^x = 0 \quad \text{--- (1)}$$

$$= x^y = y^x \quad \text{--- (2)}$$

$$\therefore \frac{dy}{dx} = \frac{-f_x}{f_y}$$

$$\frac{d}{dx}(a^x) = a^x \ln a$$

$$= a^x \cdot \ln a \cdot \frac{d}{dx}(x)$$

partially diffb (1) w.r.t "x"

$$\frac{\partial b}{\partial x} = \frac{\partial}{\partial x} (x^y - y^x)$$

$$= yx^{y-1} \cdot (1) - y^x \cdot \ln |y| \cdot (1)$$

$$= yx^{y-1} - y^x \ln y$$

from (2), $y^x = x^y$

$$= yx^{y-1} - x^y \ln y$$

$$= yx^y \cdot x^{-1} - x^y \ln y$$

$$= \frac{yx^y}{x} - x^y \ln y$$

$$= \frac{yx^y - x^y \ln y}{x}$$

$$= \frac{x^y}{x} (y - x \ln y)$$

$$= x^y \cdot x^{-1} (y - x \ln y)$$

$$b_x = x^{y-1} (y - x \ln y)$$

partially diff @ w.r.t "y"

$$\frac{\partial b}{\partial y} = \frac{\partial}{\partial y} (x^y - y^x)$$

$$= x^y \cdot \ln x / (1) - x y^{x-1} (1)$$

$$= x^y \ln x - x y^{x-1}$$

$$= x^y \ln x - x y^x \cdot y^{-1}$$

$$\therefore y^x = x^y$$

$$= x^y \ln x - x x^y \cdot y^{-1}$$

$$= x^y \ln x - \frac{x x^y}{y}$$

$$b_y = \frac{x^y \cdot y \ln x - x x^y}{y}$$

$$\frac{dy}{dx} = \frac{-b_x}{b_y} = - \left[\frac{x^{y-1} (y - x \ln y)}{x^y \cdot y \ln x - x x^y} \right]$$

$$= - \left[\frac{x^y \cdot x^{-1} \cdot y (y - x \ln y)}{x^y (y \ln x - x)} \right]$$

$$= - \left[\frac{x^{-1} \cdot y (y - x \ln y)}{y \ln x - x} \right]$$

$$\frac{dy}{dx} = \frac{y(y-x \ln y)}{x(x-y \ln x)}$$

$$\textcircled{6} \quad (\tan x)^y + y^{\cot x} = a$$

Sol:

$$b(x, y) = (\tan x)^y + y^{\cot x} - a \quad \text{--- ①}$$

$$\frac{dy}{dx} = -\frac{b_x}{b_y}$$

partially diff ① w.r.t "x"

$$\frac{\partial b}{\partial x} = \frac{\partial}{\partial x} ((\tan x)^y + y^{\cot x} - a)$$

$$= y (\tan x)^{y-1} \sec^2 x + \cancel{\cot x} (y)^{\cot x} \ln y \operatorname{cosec}^2 x$$

$$b(x) = y \tan x^{y-1} \cdot \sec^2 x + y^{\cot x} \cdot \ln y \cdot \operatorname{cosec}^2 x$$

partially diff ① from w.r.t "y"

$$\frac{\partial b}{\partial y} = \frac{\partial}{\partial y} ((\tan x)^y + y^{\cot x} - a)$$

$$= (\tan x)^y \cdot \ln \tan x + \cot x (y)^{\cot x - 1} (1)$$

$$= \tan x^y \cdot \ln \tan x + \cot x \cdot y^{\cot x - 1}$$

$$\frac{dy}{dx} = -\frac{b_x}{b_y}$$

$$\frac{dy}{dx} = - \left[\frac{y (\tan x)^{y-1} \cdot \sec^2 x - y^{\cot x} \cdot \ln y \cdot \operatorname{cosec}^2 x}{(\tan x)^y \cdot \ln(\tan x) + \cot x \cdot y^{\cot x - 1}} \right]$$

⑦

$$F(x, y, z) = 0$$

Sol:

$$F(x, y, z) = 0 \quad \text{--- ①}$$

$$\frac{dy}{dx} = \frac{-f_x}{f_y} \Rightarrow \frac{dz}{dx} = \frac{-f_x}{f_z} \Rightarrow \frac{dz}{dy} = \frac{-f_y}{f_z}$$

change it, into partial operator.
using equation ①

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} \quad ; \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$

$$\frac{\partial z}{\partial x} = -\frac{\partial F / \partial x}{\partial F / \partial z} \quad ; \quad \frac{\partial z}{\partial y} = -\frac{\partial F / \partial y}{\partial F / \partial z}$$

⑧ $b(x, y) = 0 \quad \& \quad \phi(y, z) = 0$

Sol:

$$\frac{\partial b}{\partial y} \cdot \frac{\partial \phi}{\partial z} \cdot \frac{\partial z}{\partial x} = \frac{\partial b}{\partial x} \cdot \frac{\partial \phi}{\partial y}$$

from $b(x, y)$;

$$\frac{dy}{dx} = \frac{-b_x}{b_y} \quad \text{--- ①}$$

Now, from $\phi(y, z)$;

$$\frac{dz}{dy} = -\frac{\phi_y}{\phi_x} \quad \text{--- (2)}$$

Now

$$\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx}$$

put values from (1) and (2)

$$\frac{dz}{dx} = -\frac{\phi_y}{\phi_z} \cdot -\frac{bx}{by}$$

$$by \cdot \phi_z \cdot \frac{dz}{dx} = bx \cdot \phi_y \quad \left(\begin{array}{l} -by \cdot \phi_z \cdot bx \cdot \phi_y \\ \text{are partial derivatives} \end{array} \right) \quad by = \frac{\partial}{\partial y}$$

$$\frac{\partial \phi}{\partial y} \cdot \frac{\partial \phi}{\partial z} \cdot \frac{dz}{dx} = \frac{\partial \phi}{\partial x} \cdot \frac{\partial \phi}{\partial y}$$

$$\textcircled{9} \quad x\sqrt{1-y^2} + y\sqrt{1-x^2} = a$$

Sol.

$$b(x, y) = x\sqrt{1-y^2} + y\sqrt{1-x^2} = a \quad \text{--- (1)}$$

$$= x\sqrt{1-y^2} + y\sqrt{1-x^2} - a = 0 \quad \text{--- (2)}$$

$$\frac{d^2 y}{dx^2} \Rightarrow \frac{dy}{dx} = -\frac{b_x}{b_y}$$

partially diff (2) w.r.t x:

$$b_x = \frac{\partial b}{\partial x} = \frac{\partial}{\partial x} [x\sqrt{1-y^2} + y\sqrt{1-x^2} - a]$$

$$= \sqrt{1-y^2} (1) + y \cdot \frac{1}{2} (1-x^2)^{-1/2} (-2x) - 0$$

$$= \sqrt{1-y^2} - \frac{xy}{(1-x^2)^{1/2}}$$

$$= \sqrt{1-y^2} - \frac{xy}{\sqrt{1-x^2}}$$

$$f(x) = \frac{\sqrt{1-y^2} \cdot \sqrt{1-x^2} - xy}{\sqrt{1-x^2}}$$

partially diff @ w.r.t 'y'

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (x\sqrt{1-y^2} + y\sqrt{1-x^2} - a)$$

$$= x \cdot \frac{1}{2} (1-y^2)^{-1/2} (-2y) + \sqrt{1-x^2}$$

$$= -\frac{xy}{\sqrt{1-y^2}} + \sqrt{1-x^2}$$

$$= \frac{-xy + \sqrt{1-y^2} \sqrt{1-x^2}}{\sqrt{1-y^2}}$$

$$b_y = \frac{\sqrt{1-y^2} \sqrt{1-x^2} - xy}{\sqrt{1-y^2}}$$

$$\frac{dy}{dx} = \frac{-b_x}{b_y}$$

put values

$$= \frac{-\sqrt{1-y^2}\sqrt{1-x^2}-xy}{(\sqrt{1-y^2}\sqrt{1-x^2}-xy)/\sqrt{1-y^2}}$$

$$= \frac{\sqrt{1-x^2}}{\sqrt{1-y^2}}$$

$$\frac{dy}{dx} = -\frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$$

Again simply diff w.r.t "x"

$$\frac{dy^2}{dx^2} = \frac{-\sqrt{1-x^2} \frac{d}{dx}(1-y^2)^{1/2} + \sqrt{1-y^2} \frac{d}{dx}(1-x^2)^{1/2}}{(\sqrt{1-x^2})^2}$$

$$= \frac{-\sqrt{1-x^2} \cdot \frac{1}{2}(1-y^2)^{-1/2} (2y \frac{dy}{dx}) + \sqrt{1-y^2} \cdot \frac{1}{2}(1-x^2)^{-1/2} (2x)}{(\sqrt{1-x^2})^2}$$

$$= \frac{-\sqrt{1-x^2} \cdot (1-y^2)^{-1/2} y \frac{dy}{dx} + \sqrt{1-y^2} \cdot (1-x^2)^{1/2} x}{(\sqrt{1-x^2})^2}$$

$$= \left[\frac{-y \sqrt{1-x^2} / \sqrt{1-y^2} \cdot \frac{d}{dx}}{(\sqrt{1-x^2})^2} + x \frac{\sqrt{1-y^2} / \sqrt{1-x^2}}{(\sqrt{1-x^2})^2} \right]$$

Now put values

$$= \frac{-y \frac{\sqrt{1-x^2}}{\sqrt{1-y^2}} \cdot \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}} + x \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}}{(\sqrt{1-x^2})^2}$$

$$= \frac{-y + x \sqrt{1-y^2} / \sqrt{1-x^2}}{(\sqrt{1-x^2})^2}$$

$$= \frac{y \sqrt{1-x^2} + x \sqrt{1-y^2}}{(1-x)^2 (1-x^2)^{1/2}}$$

$$\therefore y \sqrt{1-x^2} + x \sqrt{1-y^2} = a$$

$$= \frac{a}{(1-x)^2 (1-x^2)^{1/2}} = \frac{a}{(1-x^2)^{3/2}}$$

Ans (10) $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

Sol:

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad \text{--- (1)}$$

$$\frac{d^2y}{dx^2} = ? \quad \Rightarrow \quad \frac{dy}{dx} = \frac{-bx}{by}$$

partially diff w.r.t "x"

$$bx = \frac{\partial b}{\partial x} = \frac{\partial}{\partial x} (ax^2 + 2hxy + by^2 + 2gx + 2fy + c)$$

$$= 2ax + 2hy + 2g$$

$$bx = 2(ax + hy + g)$$

partially diff w.r.t "y"

$$by = \frac{\partial b}{\partial y} = \frac{\partial}{\partial y} (ax^2 + 2hxy + by^2 + 2gx + 2fy + c)$$

$$= 2hx + 2by + 2b$$

$$by = 2(hx + by + b)$$

$$\frac{dy}{dx} = - \frac{bx}{by}$$

$$= - \frac{2(ax + hy + g)}{2(hx + by + b)}$$

$$= - \frac{ax + hy + g}{hx + by + b}$$

again simple diff w.r.t "x"

$$\frac{d^2y}{dx^2} = - \left[\frac{(hx + by + b) \cdot (a + h \frac{dy}{dx}) - (ax + hy + g) \cdot (h + b \frac{dy}{dx})}{(hx + by + b)^2} \right]$$

$$= \left[\frac{ahx + h^2x \frac{dy}{dx} + aby + bh y \frac{dy}{dx} + ab + bh \frac{dy}{dx} - (ahx + ahx \frac{dy}{dx} + h^2y + h^2y \frac{dy}{dx} + gh + gb \frac{dy}{dx})}{(hx + by + b)^2} \right]$$

$$= \left[\frac{h^2x \frac{dy}{dx} + aby + ab + bh \frac{dy}{dx} - ahx \frac{dy}{dx} + h^2y + gh - gb \frac{dy}{dx}}{(hx + by + b)^2} \right]$$

$$= - \left[\frac{\frac{dy}{dx} (h^2x + abx + gb + bh) + aby + ab + h^2y + gh}{(hx + by + b)^2} \right]$$

Now put value

$$= \frac{-\left(\frac{ax+hy+g}{hx+by+b}\right)(h^2x+hb-abx-bg) + (aby+ab-h^2y-gh)}{(hx+by+b)^2}$$

$$= \frac{\left(\frac{ax+hy+g}{hx+by+b}\right)(h^2x+hb-abx-bg) - (aby+ab-h^2y-gh)}{(hx+by+b)^2}$$

$$= \frac{(ax+hy+g)(h^2x+hb-abx-bg) - (aby+ab-h^2y-gh)(hx+by+b)}{(hx+by+b)^2}$$

$$= \frac{\left[ah^2x^2 + abhx - abx^2 - abgx + h^3xy + bh^2y - abhxy - bghy + gh^2x + ghb - abgx - bg^2 - (abhxy + abhx + h^3xy + gh^2x + ab^2y + abby + bh^2y^2 + bghy + abby + ab^2 + bb^2y + bgh) \right]}{(hx+by+b)^2 \cdot (hx+by+b)}$$

$$= \frac{\left[ah^2x^2 + 2h^3xy + bh^2y^2 + 2gh^2x + 2bh^2y - abx^2 - 2abgx - 2abhxy - ab^2y^2 - 2abby - ab^2 - bg^2 + 2bgh \right]}{(hx+by+b)^3}$$

$$= \frac{\left[h^2(ax^2 + 2hxy + by^2 + 2gx + 2by) - ab(ax^2 - 2gx - 2hxy - by^2 - 2by) - ab^2 - bg^2 + 2bgh \right]}{(hx+by+b)^3}$$

$$ax^2 + 2hxy + by^2 + 2gx + 2fy = -c$$

$$= \left[\frac{h^2(-c) - ab(-c) - af^2 - bg^2 + 2bgh}{(hx + by + f)^3} \right]$$

$$\frac{d^2y}{dx^2} = \frac{abc + 2bgh - af^2 - bg^2 - ch^2}{(hx + by + f)^3}$$

⑪ $x^3 + y^3 = 3axy$

Sol:

$$x^3 + y^3 = 3axy \quad \text{--- ①}$$

$$x^3 + y^3 - 3axy = 0 \quad \text{--- ②}$$

$$\omega \frac{d^2y}{dx^2} \rightarrow \omega \frac{dxy}{dx} = \frac{-bx}{by}$$

partially diff @ w.r.t "x"

$$\frac{\partial}{\partial x} (x^3 + y^3 - 3axy)$$

$$= 3(x)^{3-1}(1) + 0 - 3ay(1)$$

$$= 3x^2 + 3ay$$

$$b(x) = 3(x^2 + ay)$$

partially diff ② w.r.t "y"

$$\frac{\partial b}{\partial y} = \frac{\partial}{\partial y} (x^3 + y^3 - 3axy)$$

$$= 3y^2 - 3ax$$

$$by = 3(y^2 - ax)$$

$$\frac{dy}{dx} = \frac{-b(x)}{by}$$

$$\frac{dx}{dy} = - \left[\frac{3(x^2 - ay)}{3(y^2 - ax)} \right]$$

$$= \frac{-x^2 + ay}{y^2 - ax} \Rightarrow \frac{ay - x^2}{y^2 - ax}$$

again simple diff w.r.t "x"

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{ay - x^2}{y^2 - ax} \right)$$

$$= \frac{y^2 - ax \frac{d}{dx}(ay - x^2) - (ay - x^2) \frac{d}{dx}(y^2 - ax)}{(y^2 - ax)^2}$$

$$= \frac{(y^2 - ax) \left(a \frac{dy}{dx} - 2x \right) - (ay - x^2) \left(2y \frac{dy}{dx} - a \right)}{(y^2 - ax)^2}$$

$$= \frac{\left[y^2 a \frac{dy}{dx} - 2xy^2 - a^2 x \frac{dy}{dx} + yax^2 - (2ay^2 \frac{dy}{dx} + ay^2 + 2x^2 y \frac{dy}{dx} - ax^2) \right]}{(y^2 - ax)^2}$$

$$= \frac{-2xy^2 - a^2x \frac{dy}{dx} + ax^2 - ay^2 \frac{dy}{dx} + a^2y + 2x^2y \frac{dy}{dx}}{(y^2 - ax)^2}$$

$$= \left(\frac{\frac{dy}{dx} (-a^2x - ay^2 + 2x^2y) - (2xy^2 + ax^2 + a^2y)}{(y^2 - ax)^2} \right)$$

Now put value $\frac{dy}{dx}$

$$= \left(\frac{\left[\frac{ay - x^2}{y^2 - ax} \right] (2x^2y - ay^2 - a^2x) - (2xy^2 + ax^2 + a^2y)}{(y^2 - ax)^2} \right)$$

$$= \left(\frac{(ay - x^2)(2x^2y - ay^2 - a^2x) - (y^2 - ax)(2xy^2 + ax^2 + a^2y)}{(y^2 - ax)^2 (y^2 - ax)} \right)$$

$$= \left(\frac{2ax^2y^2 - a^2y^3 - a^3xy - 2x^4y + ax^2y^2 + a^2x^3 - 2xy^4 + ax^2y^2 + a^2y^3 + 2ax^2y + a^2x^3 + a^3xy}{(y^2 - ax)^2 (y^2 - ax)} \right)$$

$$= \frac{6ax^2y^2 - 2a^3xy - 2xy(x^3 + y^3)}{(y^2 - ax)^3}$$

$$= \frac{6ax^2y^2 - 2a^3xy - 2xy(3axy)}{(y^2 - ax)^3}$$

$$= \frac{6ax^2y^2 - 2a^3xy - 6ax^2y^2}{(y^2 - ax)^3}$$

$$= \frac{-2a^3ny}{(y^2-ax)^3}$$

$$= \frac{+2a^3ny}{+1(ax-y^2)^3}$$

$$\frac{d^2y}{dx^2} = \frac{2a^3ny}{(ax-y^2)^3}$$

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